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PDA Research

2019 PDA Traceability of Primary Packaging Survey

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PDA Research

2019 Traceability of Primary Packaging Survey

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PDA conducts benchmarking surveys to assess the state of the industries we serve. We explore such topics as the current challenges in regulatory science, quality, analytical and manufacturing site operations, and processes of pharmaceuticals and biotechnology products.

These surveys are research tools used to gain information, usually from a targeted audience within PDA's areas of interest.

The results, which are published or presented at meetings, represent the perspective of PDA survey respondents. Findings are used to focus PDA support activities or to contribute to the development of technical documents, courses, and meetings, fulfilling the professional needs and bringing additional value to PDA members.

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Introduction

Serialization in the pharmaceutical supply chain is a growing effort to improve shortcomings within the current track and trace processes. During the last Prefilled Syringes Interest Group meeting at the PDA Packaging Conference, March 2019, in Venice; there was an introduction to an initiative for marking of containers with a unique identifier [code]. The overall concept is shown in the picture below:

Figure 1 Identification Code on Primary Packaging

Currently there are global regulatory mandates for the serialization and traceability of secondary packaging. Some countries are considering extending these mandates to primary containers and thus understanding the capabilities will enable the PDA and members to engage with governments and regulatory agencies with a baseline knowledge of the current industry views on the need for traceability in primary packaging.

This survey was intended to learn more about how companies which manufacture parental drug products are working to improve tracking efforts and if serialization is the future for primary packaging components. This survey was designed to be completed by individuals within companies that are involved in the manufacture, development, and handling of drug products or ingredients. The survey will help PDA members, industry, and regulators understand the current state of demand and issues surrounding track and trace, as well as attaining additional insight to the discussions that may be occurring at the site level to improve controls and traceability of drug product from production to the patient and intern patient safety.

The current challenges for the industry are related to increasing product complexity, lower volumes and higher flexibility requirements along with increased regulatory scrutiny on product traceability. USP 1790 is a guidance document for the inspection of prefilled syringes. It is now not enough to inspect and reject products. Manufacturers now have the responsibility to determine the normal reject rate and in case of a deviation or trend to investigate for the root causes using Statistical Process Control [SPC] principles and apply corrective actions and validated alert levels. The traceability of primary containers is a risk reduction tool that provides single product insight into the inspection process, enabling a detailed corrective actions and risk reduction strategies.

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Methodology

The survey was open to the general public for participation from September 25 to November 8th, 2019. The preliminary results were aggregated and will be presented at the PDA Packaging Conference February 25-26, 2020 in Basel Switzerland. The full results will be published as PDA Research and will support and future PDA initiatives pertaining to supply chain, serialization and traceability, as appropriate. The results will be aggregated in a manner designed to protect the confidentiality of the gathered information and data.

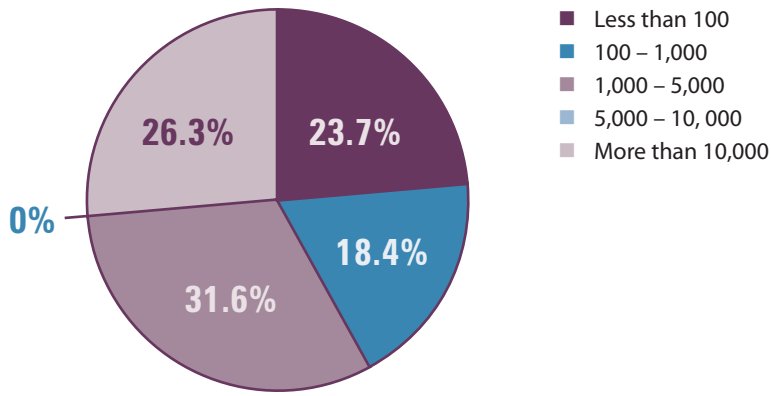
The 2019 Container Traceability Survey is based on the views of 38 industry professionals across North America, Latin America, Europe, Asia/Pacific, Africa, and the Middle East. Survey respondents are involved in manufacture and packaging activities within their current companies who possess an understanding of their companies' procedures and needs.

Demographics (Questions 1-3)

Many organizations manufacture, develop, or contract manufacture parenteral drug products across a range of drug types. The following survey questions queried the demographics of respondents to assess representation of the pharmaceutical and biopharmaceutical manufacturing industries. The responses to the following questions demonstrate that the survey represents a broad span of industry with a good distribution of small and large companies, multiple product types, and global reach **(Questions 1-3).**

1. What is the size of your company (number of employees)?

Answer	Count	%
Less than 100	9	23.7%
100 – 1,000	7	18.4%
1,000 – 5,000	10	31.6%
5,000 – 10, 000	0	0%
More than 10,000	12	26.3%
	Answered	***38***



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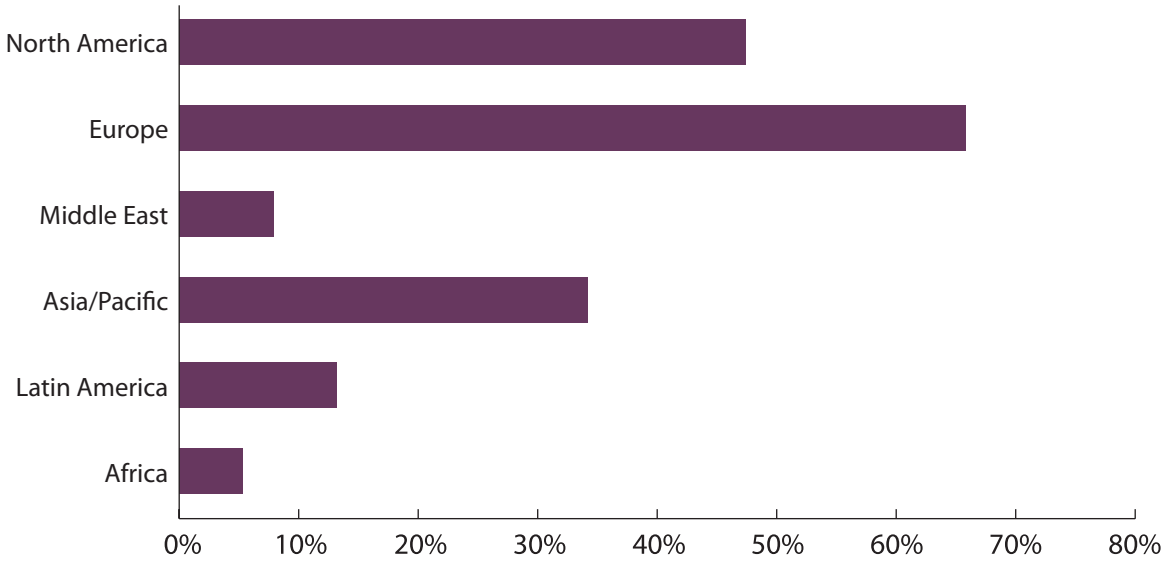
2. Where are your company's internal development/manufacturing locations? (please check all that apply)

Answer	Count	%
North America	18	47.4%
Europe	25	65.8%
Middle East	3	7.9%
Asia/Pacific	13	34.2%
Latin America	5	13.2%
Africa	2	5.3%
	Answered	***38***



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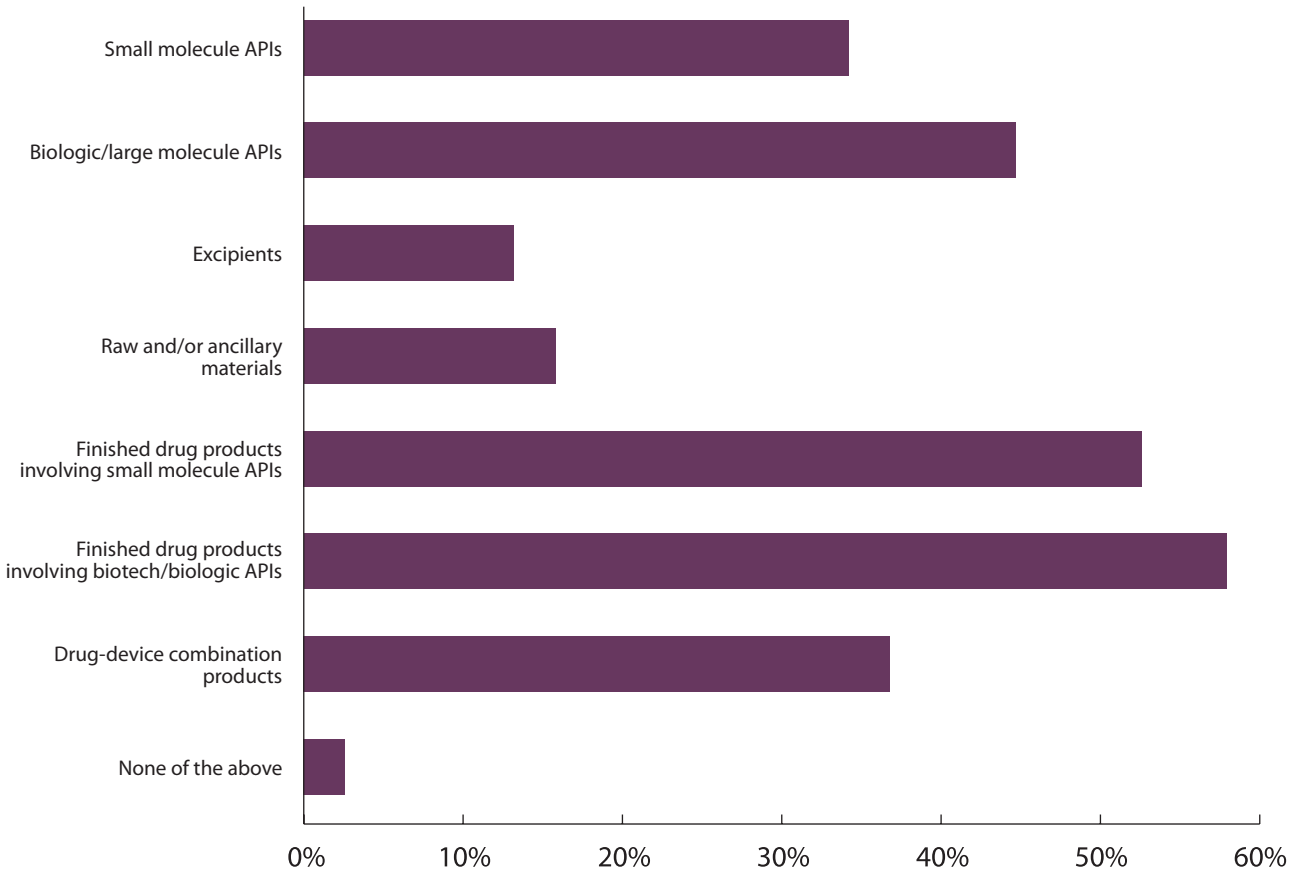
3. What types of products does your company manufacture, develop, or contract for manufacturing? (please check all that apply)

Answer	Count	%
Small molecule APIs	13	34.2%
Biologic/large molecule APIs	17	44.7%
Excipients	5	13.2%
Raw and/or ancillary materials	6	15.8%
Finished drug products involving small molecule APIs	20	52.6%
Finished drug products involving biotech/biologic APIs	22	57.9%
Drug-device combination products	14	36.8%
None of the above	1	2.6%
	Answered	***38***



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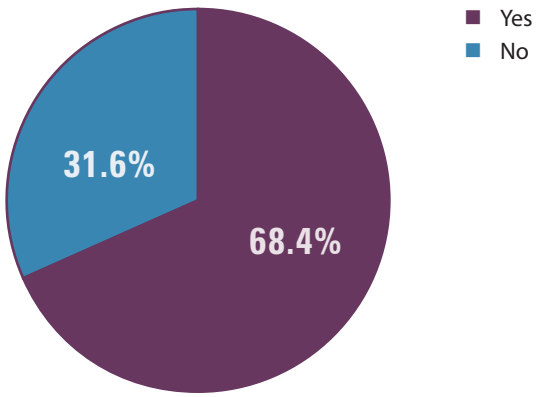
Container Traceability (Questions 4-12)

The initial survey questions queried the demographics of respondents to assess representation of the pharmaceutical and biopharmaceutical industries. The responses to the following questions demonstrate the current industry perspective for those who manufacture, develop, or contract manufacture parenteral drug products and are involved in the packaging process **(Questions 4-12).**

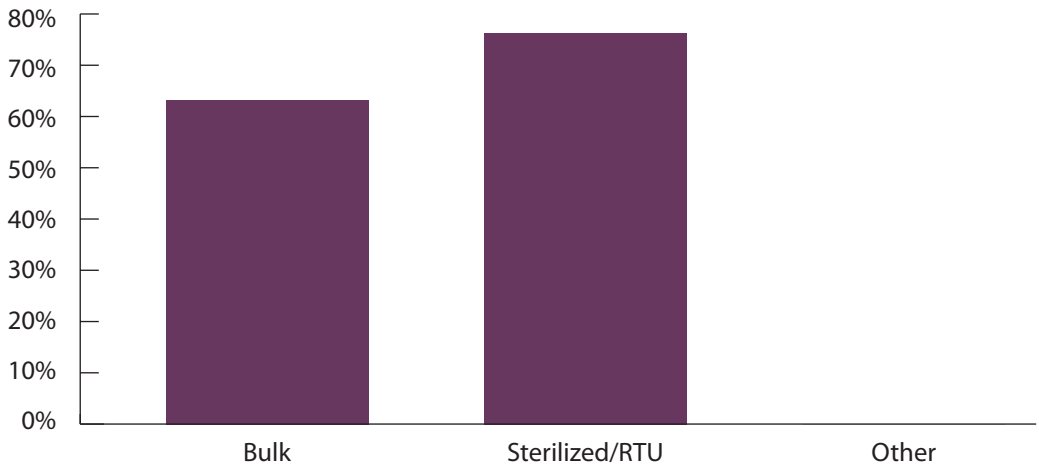
4. Have you considered marking the primary container to facilitate internal tracking or reconciliation on your filling lines, inspection, or secondary packaging lines?

Answer	Count	%
Yes	26	68.4%
No	12	31.6%
	Answered	***38***



5. Do you handle bulk product or pre-sterilized [Ready to Use (RTU)] products?

Answer	Count	%
Bulk	24	63.2%
Sterilized/RTU	29	76.3%
Other (please specify)	0	0%
	Answered	***38***



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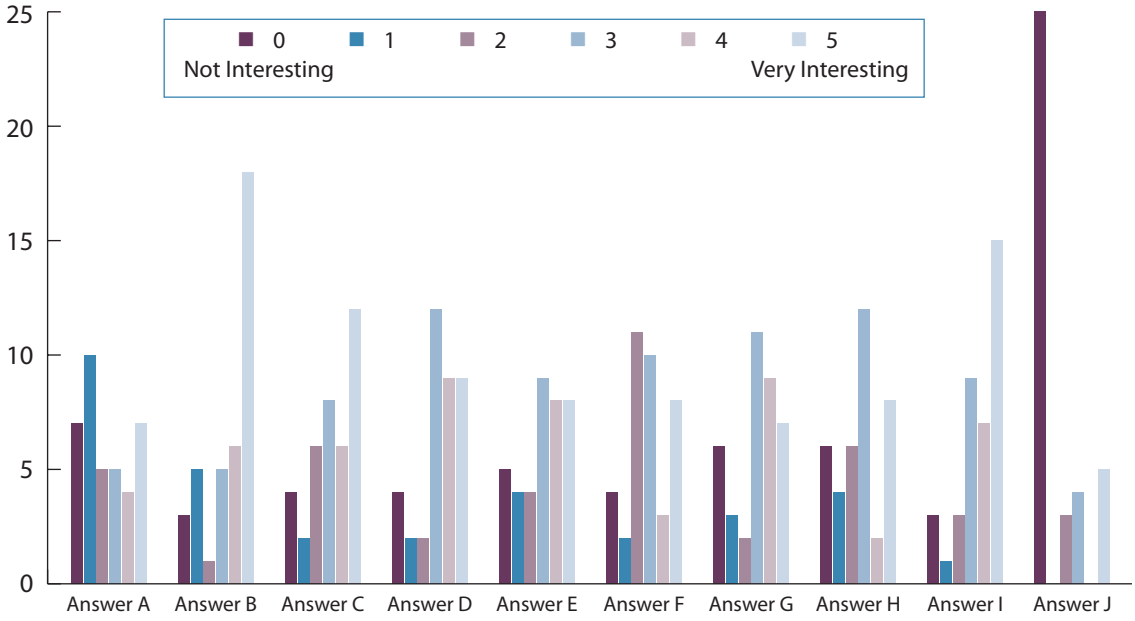
6. On a scale of 0–5, what are the greatest considerations to institute/ implement traceability of primary containers and to build a business case? ("0" – not interesting; "5" – very interesting)

Answer	0	1	2	3	4	5
A. Mix-up avoidance of incoming containers	7	10	5	5	4	7
B. Mix-up avoidance of filled containers	3	5	1	5	6	18
C. To automate and simplify reconciliation on filling lines at container level	4	2	6	8	6	12
D. Reducing risk through time limited batch segregation in the case of quality issues	4	2	2	12	9	9
E. Reject tracking of filled containers within the automated inspection processes	5	4	4	9	8	8
F. Overall improvement of manufacturing process and efficiency (i.e., OEE, TCO etc.)	4	2	11	10	3	8
G. Segregation of problematic marketed drugs due to primary containers and / or filled drug product	6	3	2	11	9	7
H. Improved clinical trial monitoring at each individual patient level	6	4	6	12	2	8
I. Better traceability in case of product recall	3	1	3	9	7	15
J. Additional considerations, not included above (please specify)	25	0	3	4	0	5
Answered	***38***					

Additional considerations, not included above (please specify):

- Automated line clearance ("5")
- Avoids spurious products ("3")
- Costs ("2")
- Customer confident on the product ("5")
- Regulations ("5")
- Support serialization integrity ("3")
- Tracing of manufacture/ transport/ storage history and use ("5")
- Using it as Anti-Counterfeiting feature at primary packaging level ("5")
- None ("0")



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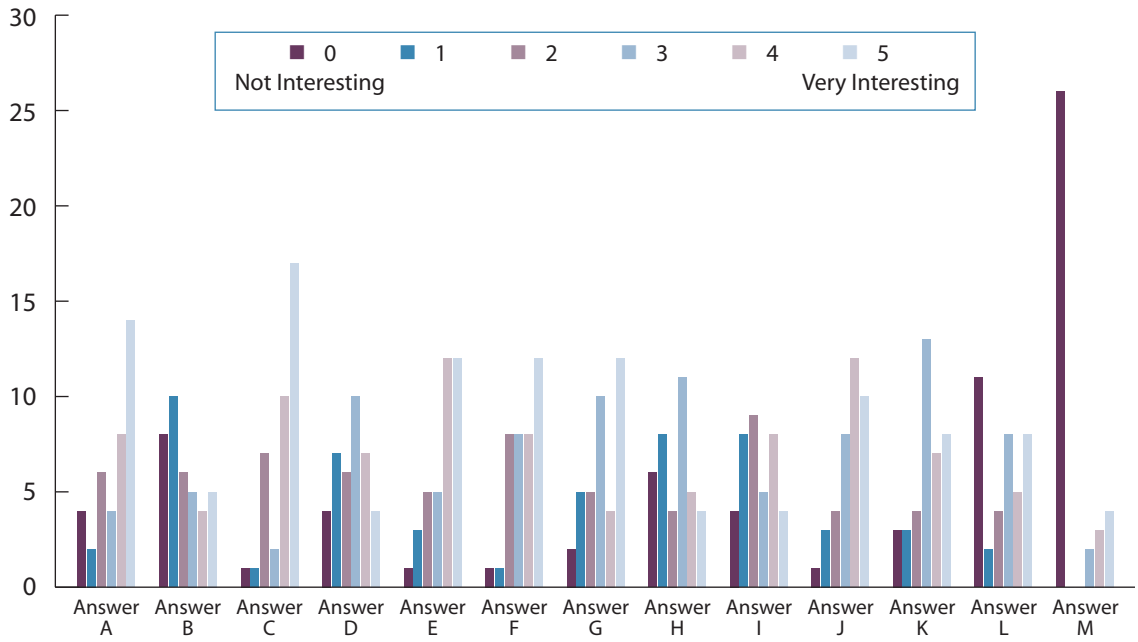
7. On a scale of 0-5, what do you believe are the potential roadblocks in using and managing serialized primary containers internally? ("0" – not interesting; "5" – very interesting)

Answer	0	1	2	3	4	5
A. A lack of accepted standards	4	2	6	4	8	14
B. Negative perception of the regulators - FDA, etc.	8	10	6	5	4	5
C. Traceability implementation on existing filling lines	1	1	7	2	10	17
D. Traceability implementation on new filling lines	4	7	6	10	7	4
E. Upgrading existing automated inspection machines	1	3	5	5	12	12
F. Product availability - number of vendors	1	1	8	8	8	12
G. Creating a business case	2	5	5	10	4	12
H. Segregation of problematic marketed drugs due to containers	6	8	4	11	5	4
I. Process validation	4	8	9	5	8	4
J. Data management and systems integration	1	3	4	8	12	10
K. Implementation support	3	3	4	13	7	8
L. Other priorities	11	2	4	8	5	8
M. Additional challenges, not included above (please specify)	26	0	0	2	3	4
Answered	***38***					

Additional considerations, not included above (please specify):

- Technical feasibility and limits (product vs code size, throughput rate, OEE impact) ("5")
- Only makes sense if an overall industry standard is generated ("5")
- Training, specific testing methods, AQL limits ("4")
- Economical repercussion ("5")
- Data privacy regulations ("4")
- Beta test ("3")
- The shelf life of the 2d code ("4")
- None ("0")



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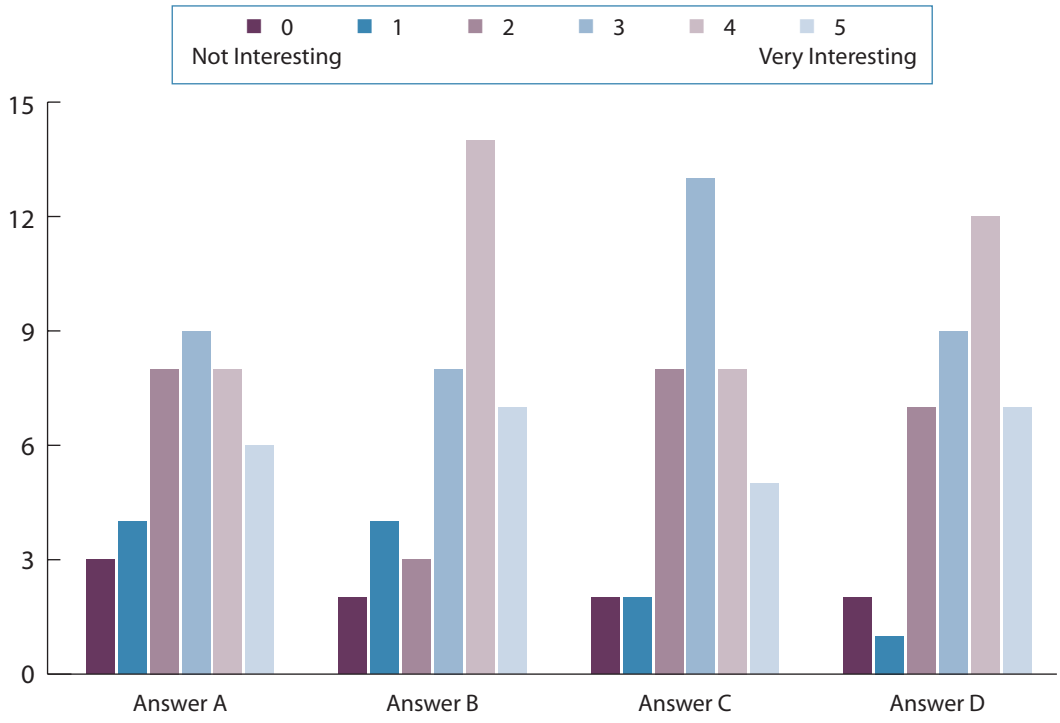
8. On a scale of 0-5, do you believe that the considerations listed below are potential roadblocks in using and managing serialized primary containers internally? ("0" – not interesting; "5" – very interesting)

Answer	0	1	2	3	4	5
A. Ability of existing automated inspection systems to read 2D barcodes	3	4	8	9	8	6
B. Addition of codes may impact effectiveness and efficiency of automated inspection systems	2	4	3	8	14	7
C. PFS (Prefilled Syringes) safety system may impact readability of codes	2	2	8	13	8	5
D. Adding a code may have a regulatory impact for the product	2	1	7	9	12	7
Answered	***38***					



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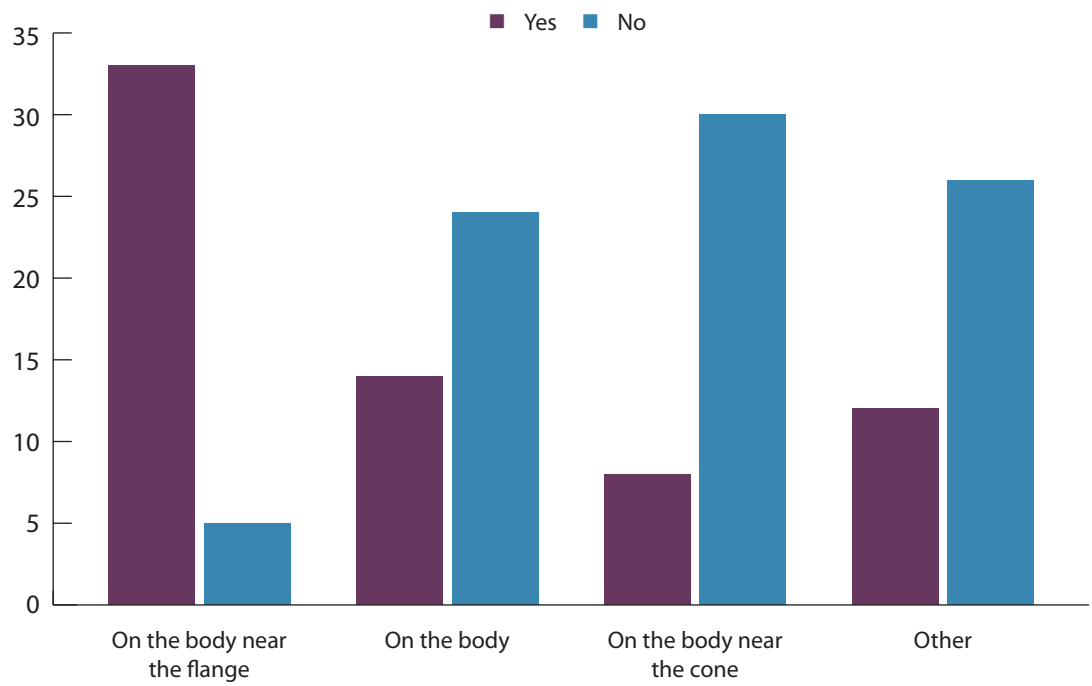
9. In your opinion, the best position on the syringe (see Figure 1) for the machine-readable 2D barcode containing unique code is?

Answer	Yes	No
On the body near the flange	33	5
On the body	14	24
On the body near the cone	8	30
Other (please specify)	12	26
	Answered	***38***
	Skipped	

Other (please specify):

- All
- Flange
- On finger grips
- At the flange
- On the flange
- Flange top for easy readability
- Readability may be issue if it is near the cone
- On secondary packaging material
- None



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For Questions 10–11 there were no options provided, only respondent feedback.

10. Are there any product/process/validation considerations that have not been covered in this survey and should be highlighted?

Answer:

- Cold Chain Product.
- How to consider the codes of material losses and wastage of primary packing material.
- Interference with labeling, label readability and readability of the matrix after final labeling of the container.
- Introduction of system to read the printed code/ in terms of marketed product what will change for the final user? Final user will read the serialization on the final box and no on the single vial/syringe. So the serialization code on the Primary packing will be used mainly for internal company purpose.
- Many syringes are hold by the body using vacuum wheels or star-wheels. To read a code on the body requires huge modifications. The goal should be, to have the code on the flange to read from the top.
- Primary and Secondary packaging links.
- Problematic on tracking when have the vial being filled to investigate deviations.
- Should met the Standard Primary Packaging Requirements like USP660 and Annex 1.
- No additional input based on my experience.

Answered **9**

11. Are there any specific requirements you would have for the primary container that have not been covered in this survey and should be highlighted?

Answer:

- Authenticity and disposal control at every level till end user and re-entering of used container by Mis-branded people.
- Coding requirements shall follow strictly an international coding standard (like GS1 for instance).
- Determine any impact on glass strength.
- Expand availability of ready to use primary pck.
- I think the necessity in case of becoming real must be assume by the primary packaging materials manufacturers.
- RFID

Answered **6**

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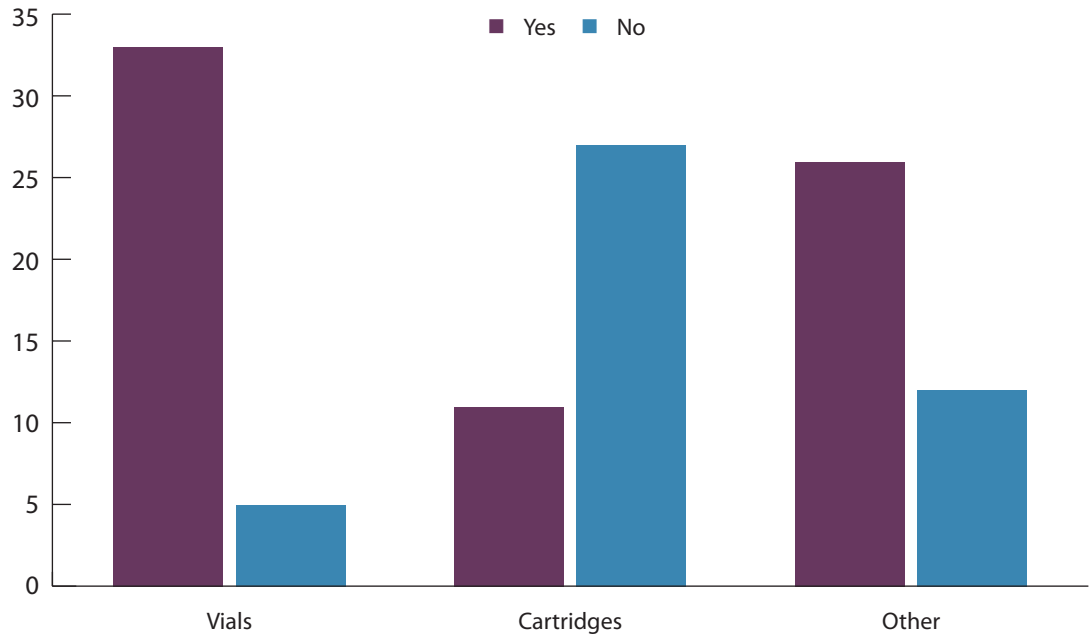
12. Do you think that primary container traceability can be applied to other containers used by your company?

Answer	Yes	No
Vials	33	5
Cartridges	11	27
Other (please specify)	26	12
	Answered	***38***
	Skipped	

Other (please specify):

- Metal seal of vial
- It can be applied on needle protecting devices both passive and active
- Ampoules
- Pre filled syringe
- None



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Conclusion

The organizations that participated in the survey develop, manufacture, develop, or contract manufacture parental drug products. The initial survey questions queried the demographics of respondents to assess representation of the pharmaceutical and biopharmaceutical industries. The serialization of primary packaging could potentially improve manufacturing efficiency or total cost of ownership and quality process. Typically, this would cover:

1. Reducing risk through time limited batch segregation in the case of quality issues
2. Mix-up avoidance of filled containers
3. Reject cause analysis in fillingeject tracking and traceability of filled containers within the automated inspection processes
4. Traceability of product to packaging

This survey enabled the respondents to understand industry needs and opportunities for improvement. Despite the modest number of completed responses, the participants do represent a comprehensive span of industry with a good distribution of small and large companies, multiple product types, and a global representation. There is interest in the primary container serialization and the perceived initial value in the serialization of primary packaging components. Interestingly, additional areas for adoption were also proposed.

While the survey was initially developed and discussed at the PDA Pre-filled Syringe Interest Group, there was also interest from respondents on potential use of sterilization for container traceability with other product types such as vials and cartridges.

The responses have also shown that there is a need to develop certain areas of serialization and further support a commonality of approaches across pharmaceutical companies, vendors and equipment providers. This survey will support PDA, industry, and regulators in understanding the current state of the industry, as well as getting insights to the discussions that may be occurring at the site level to enhance traceability and manufacturing operations.

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PDA Survey: *2019 Traceability of Primary Packaging Survey*

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